Kotlin Packages

Kotlin Packages (Namespace)

Namespace

A **namespace** is a logical container that groups related identifiers such as classes, functions, and variables to avoid naming conflicts. It provides a way to organize code systematically, ensuring that different components with the same name do not interfere with each other.

In Kotlin, **packages act as namespaces**, helping developers manage code efficiently. Every class or function within a package belongs to its **namespace**, preventing conflicts with similar names in other packages.

Kotlin Packages (Namespace)

Uses of Packages

1. **Code Organization** – Helps structure code by grouping related components together.
2. **Avoiding Name Conflicts** – Prevents class name duplication by providing unique namespaces.
3. **Modularity and Reusability** – Encourages modular programming, making code reusable across different parts of an application.
4. **Improved Maintainability** – Simplifies debugging and updates by organizing code logically.

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5. Access Control – Packages help manage visibility and accessibility of classes and functions.

6. Efficient Collaboration – Allows multiple developers to work on different modules independently.

7. Integration with Android Framework – Aligns with Android's structure, ensuring smooth integration with system components.

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Package Declaration: Every Kotlin file in an Android project starts with a package declaration. If omitted, it defaults to the root package.

package com.example.myapp

Folder Structure:

- The **package name** follows the folder structure inside src/main/java or src/main/kotlin.
- Example: If a class is inside src/main/java/com/example/myapp/utils, its package would be: package com.example.myapp.utils

Kotlin Packages

- **Accessing Classes from Other Packages:**
  - You can import classes from other packages using the import statement.
  - Example: import com.example.myapp.utils.MyHelper
- **Default Android Package:**
  - The main package in an Android project is usually the **application ID** (e.g., com.example.myapp).
  - This package is defined in the AndroidManifest.xml file.
- **Custom Packages:** You can create custom packages (e.g., utils, network, models, etc.) to organize code better.